THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

IKANA SECONDARY SCHOOL
FORM THREE MID – TERM EXAMINATION

033/2A **BIOLOGY 2A**

Time: 2Hours **August, 2025**

INSTRUCTIONS

1. This paper consists of two questions.
2. Answer **all** questions.
3. All writings should be in blue or black pen, **except** for diagrams that must be drawn in pencil.
4. All communication devices, and any unauthorized materials are **NOT** allowed in the an examination room.
5. Write your **Name** on every page of your answer sheets.

1. You have been provided with solid sample **T**, two beakers labeled **A** and **B**, measuring cylinder, spatula, hot and cold water. Fill beaker **A** and **B** with 80cm^3 of hot water and cold water respectively. Using a spatula, slowly put a small portion of the solid sample **T** in a beaker **A** and **B** respectively, and observe the changes in both beakers for 5 minutes, then answer the questions that follow.
   (a) (i) Explain what happened in beaker **A** and **B**.
   (ii) What is the difference in observation between beaker **A** and **B**.
   (iii) Give a reason for the difference in a(ii).
   (b) What is the aim of the experiment?
   (c) (i) Name the physiological process that is investigated in this experiment.
   (ii) Give the meaning of the process you named in c(i).
   (iii) How is the process named in c(i) significant to animals and plants (4 points)
   (d) Write down three organs found in human and two organs found in plants that perform their work by the process mentioned in c(i) above.
   (e) Explain any two factors that affect the rate of the process mentioned in c(i) above.

2. You have been provided with specimen **U**, **V**, **W**, **X** and **Y.** Observe them carefully then answer the questions that follow.
   (a) Identify specimen **U, V, W, X** and **Y** using their common names,
   (b) Classify specimen **W, X** and **Y** into their classes.
   (c)Why it is important to do the action in (b) above? (2 points)
   (d) (i) Draw a well labeled diagram of specimen **V.**
   (ii) Name the process which occurs in specimen **V** during day time.
   (iii) What are the two products produced by the process mentioned in d(ii) above?
   (iv) State any three adaptations of specimen **V** to the process in d(ii) above.
   (e)How is the kingdom to which specimen **U** belongs is significant to development of industries? (3 points)